STATEMENT OF TECHNICAL REQUIREMENTS

Navigation Radar and ECDIS System

Project	OSV-45 Multipurpose Offshore Support Vessel (Hull 101)
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Revision	A
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Prepared by	Example Shipbuilders Ltd.

Vendors shall record **Complied** or **Not Complied** against every clause, with remarks for any deviation.

Chapter 01 - General Information

Clause	Requirement	Vendor reply
1.1	This Statement of Technical Requirements defines the requirements for the supply of a navigation radar and ECDIS system for the vessel.	
1.2	The scope covers design, manufacture, type testing, supply, installation supervision, commissioning, training and product support.	

Chapter 02 - Operational Requirements

Clause	Requirement	Vendor reply
2.1	The system shall comprise one X-band radar, one S-band radar and two ECDIS workstations networked on a redundant bridge LAN.	
2.2	Each workstation shall be able to act as a radar display or an ECDIS display, selectable by the operator.	
2.3	The radar shall provide automatic target tracking of not less than 100 targets and AIS target display.	
2.4	The ECDIS shall hold a complete back-up of chart data, and failure of one workstation shall not affect the other.	

Chapter 03 - Technical Specifications

Clause	Requirement	Vendor reply
3.1	The X-band radar shall have a transmitter output of not less than 25 kW and the S-band radar not less than 30 kW.	
3.2	Display size shall be not less than 23.1 inches (diagonal) with a resolution of not less than 1920 x 1200 pixels.	
3.3	The radar shall comply with IMO Resolution MSC.192(79) and IEC 62388. The ECDIS shall comply with IMO Resolution MSC.232(82) and IEC 61174.	
3.4	Interfaces to gyro, DGPS, AIS, EM log, echo sounder and VDR shall conform to IEC 61162-1/-2 and shall be included in the base price.	
3.5	Power supply shall be 230 V AC, 50 Hz from the ship's mains with a dedicated UPS giving not less than 30 minutes of operation on battery.	

Chapter 04 - Environmental Specifications

Clause	Requirement	Vendor reply
4.1	All equipment shall be type tested to IEC 60945 for the applicable equipment category.	
4.2	Antenna units shall have a degree of protection not less than IP56 and operate from -25 C to +55 C.	
4.3	Below-deck units shall operate from -15 C to +55 C at up to 95% relative humidity.	

Chapter 05 - Reliability and Maintainability

Clause	Requirement	Vendor reply
5.1	The vendor shall state the MTBF of each major unit, which shall be not less than 10,000 hours, and the MTTR, which shall not exceed 2 hours.	
5.2	Built-in test equipment shall identify a failed unit down to line-replaceable-unit level.	
5.3	Product support, including spares availability, shall be guaranteed for not less than ten (10) years after delivery.	

Chapter 06 - General Requirements

Clause	Requirement	Vendor reply
6.1	The equipment shall carry a valid marine equipment type-approval certificate; copies shall accompany the offer.	
6.2	On-board spares for two (2) years of operation shall be included. A priced list of recommended base spares shall be submitted.	
6.3	The vendor shall conduct FAT, and shall attend HAT and SAT on board at no extra cost.	
6.4	On-board training shall be provided for not less than 6 operators and 2 maintainers.	
6.5	The warranty shall be not less than twenty-four (24) months from vessel delivery.	
6.6	Manuals, drawings and test records shall be delivered in English, in printed and searchable electronic form.	
6.7	Delivery to the yard shall be within eight (8) months of the purchase order.	